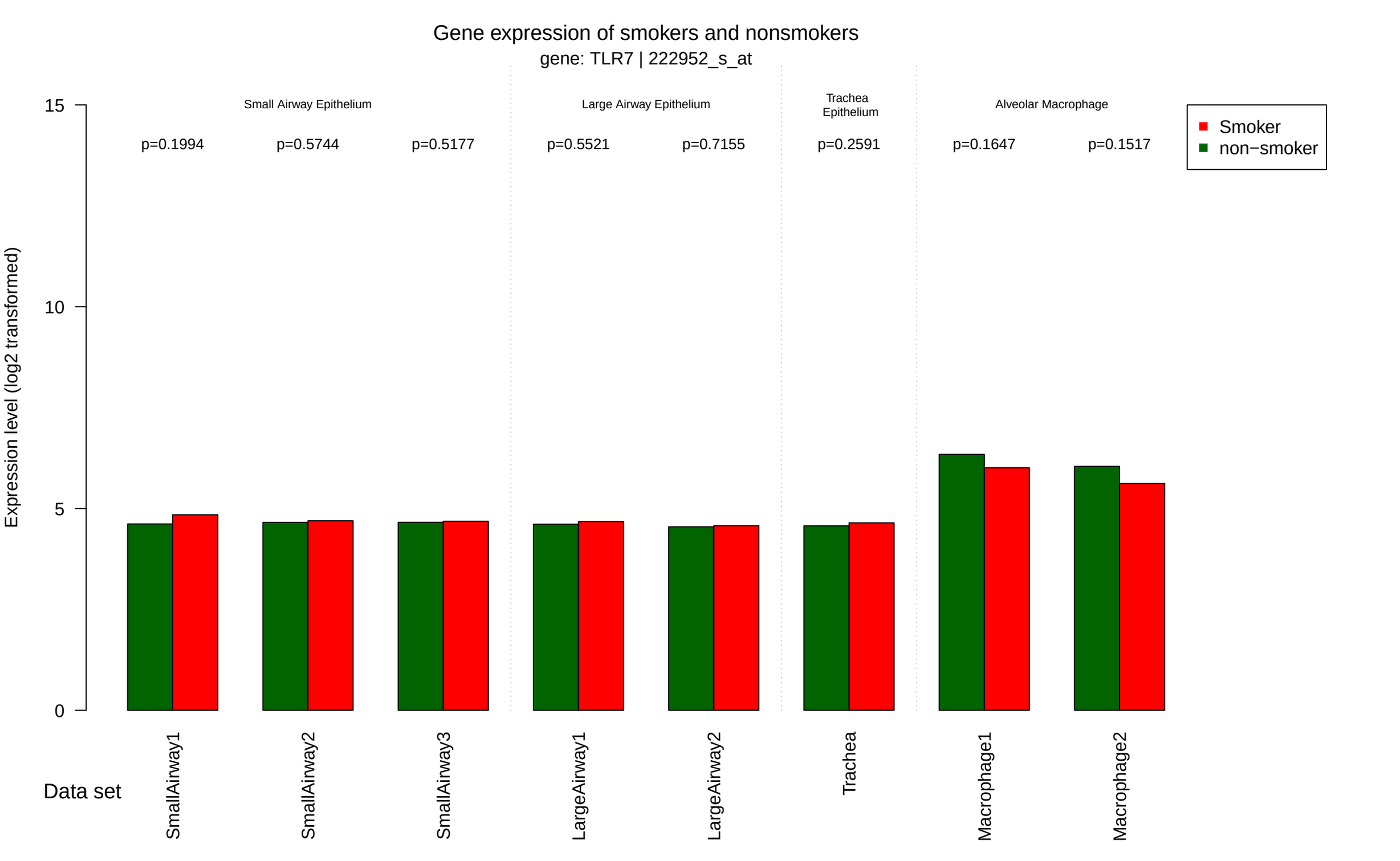
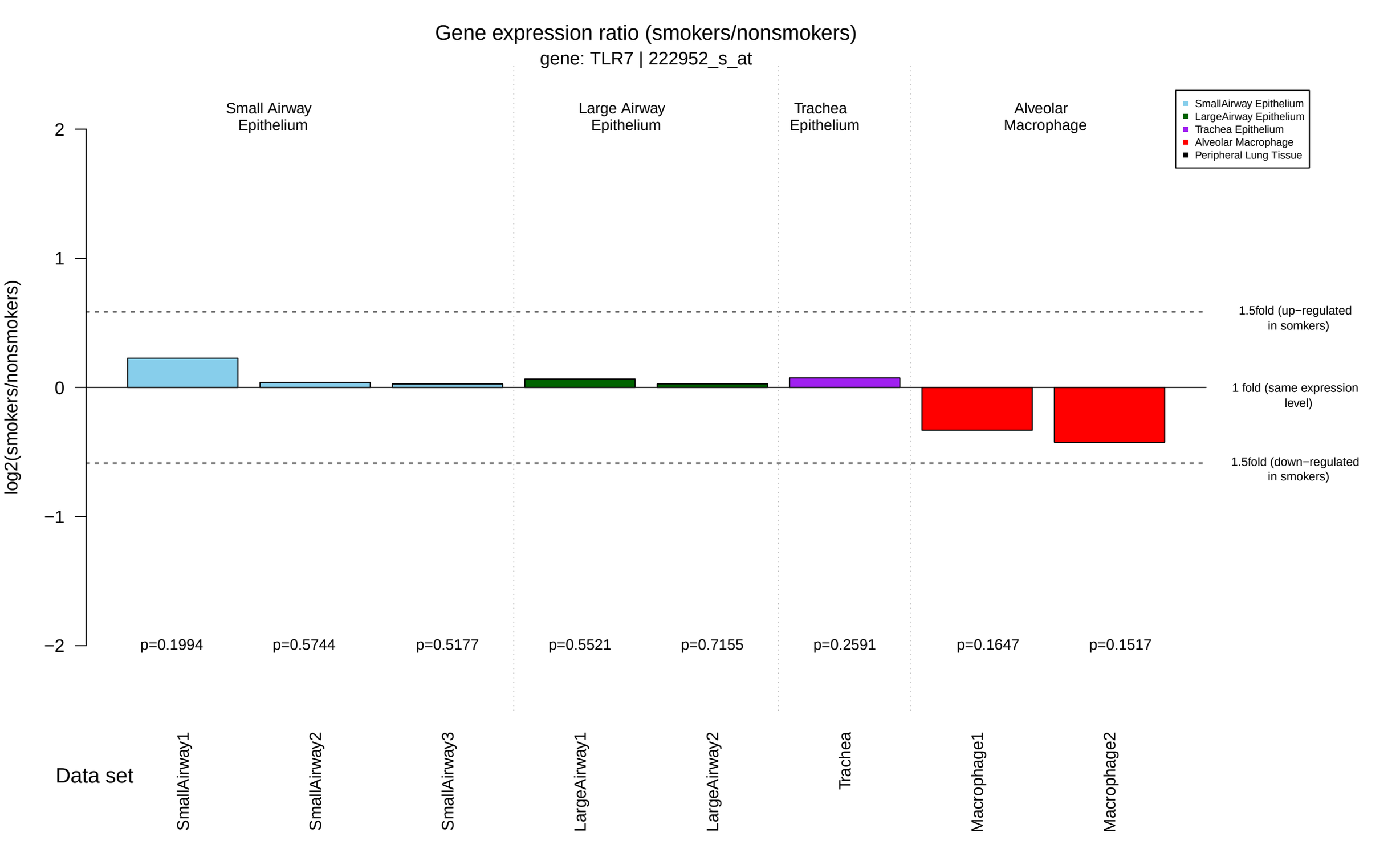
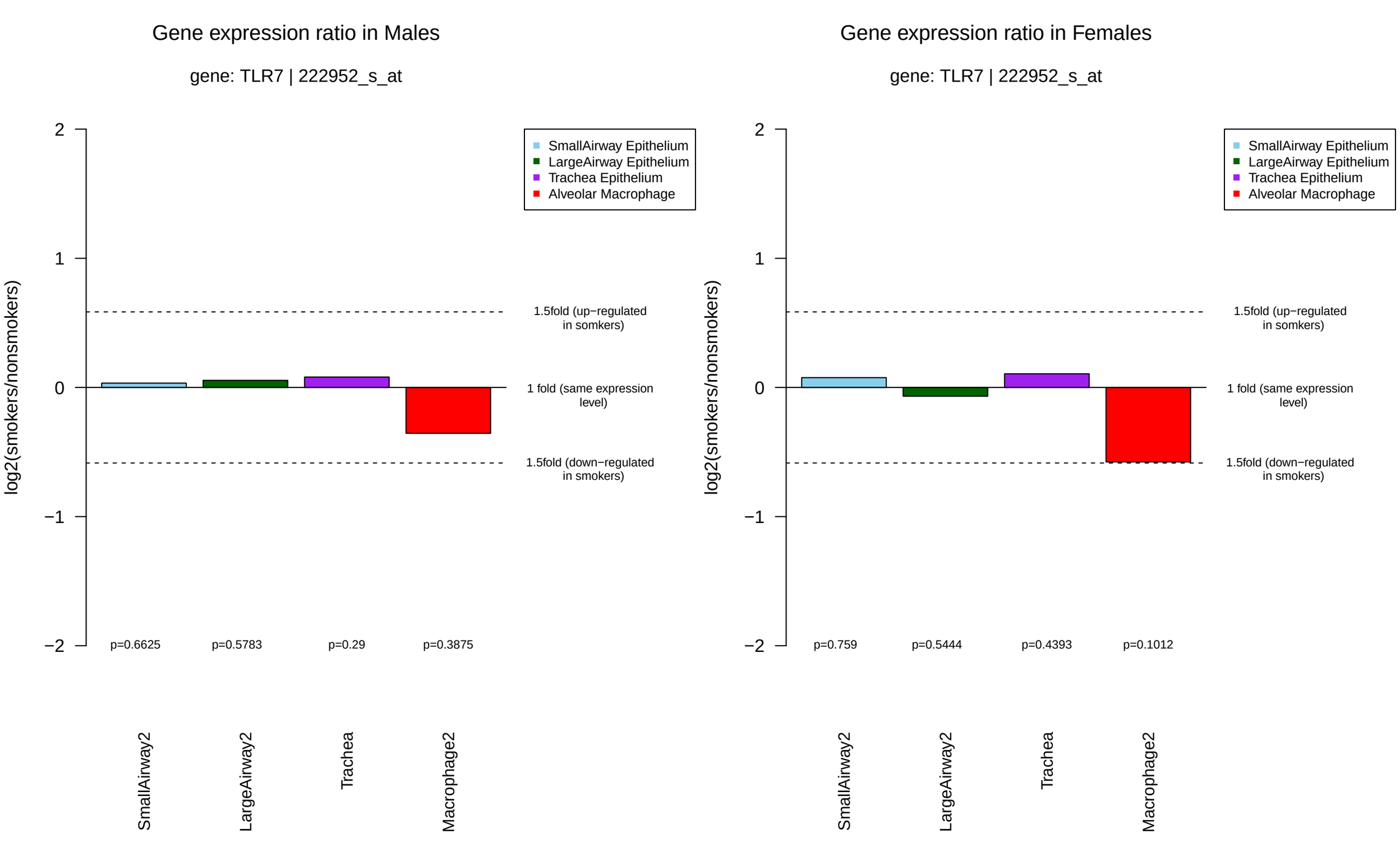
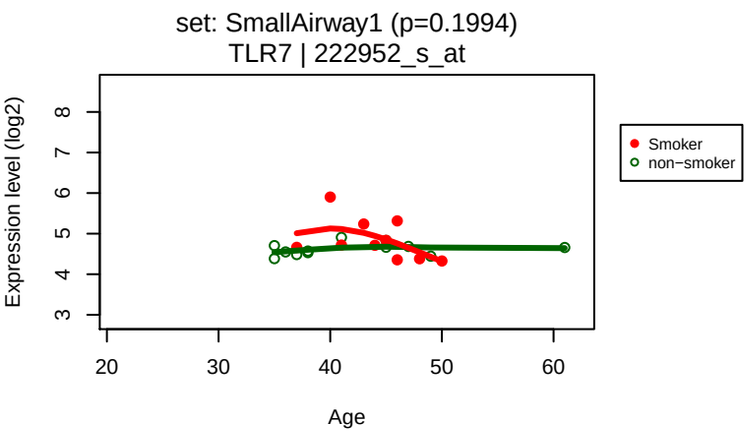
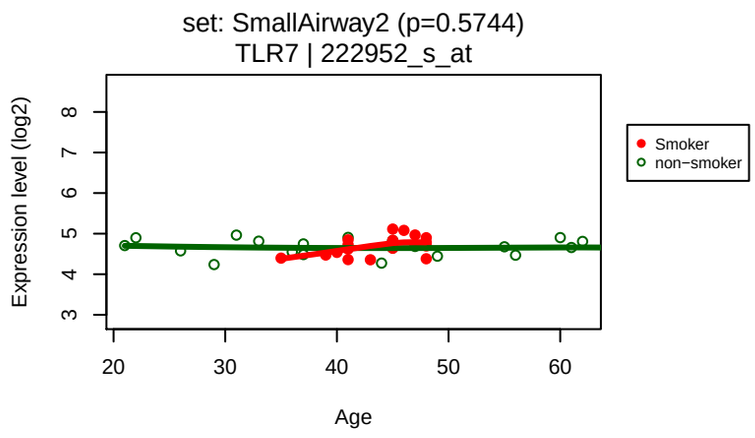
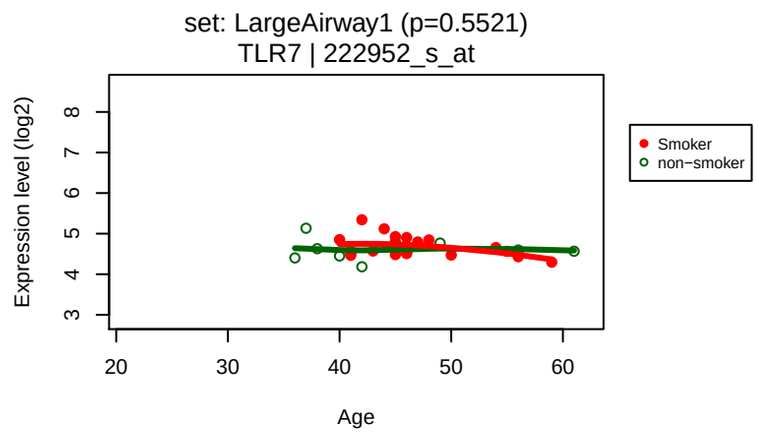

Gene expression of smokers and nonsmokers
gene: TLR7 | 222952_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1994
p=0.5744
p=0.5177
p=0.5521
p=0.7155
p=0.2591
p=0.1647
p=0.1517
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

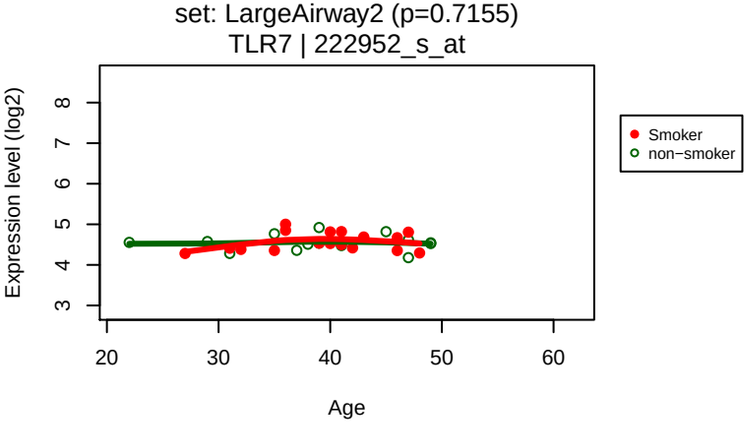

Gene expression ratio (smokers/nonsmokers)
gene: TLR7 | 222952_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1994
p=0.5744
p=0.5177
p=0.5521
p=0.7155
p=0.2591
p=0.1647
p=0.1517
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

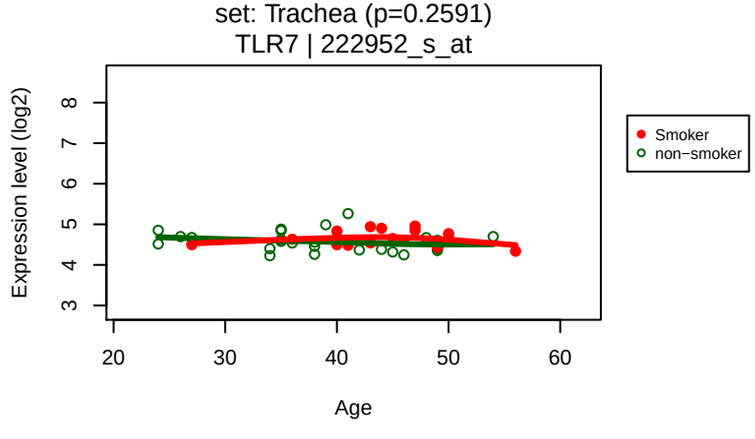

Gene expression ratio in Males
gene: TLR7 | 222952_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6625
p=0.5783
p=0.29
p=0.3875
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TLR7 | 222952_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.759
p=0.5444
p=0.4393
p=0.1012
SmallAirway2
LargeAirway2
Trachea
Macrophage2

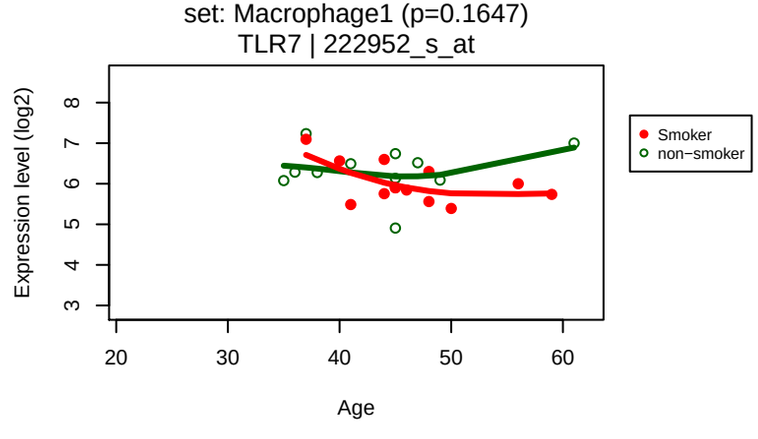

set: SmallAirway1 (p=0.1994)
TLR7 | 222952_s_at
Expression level (log2)
Age
Smoker
non−smoker

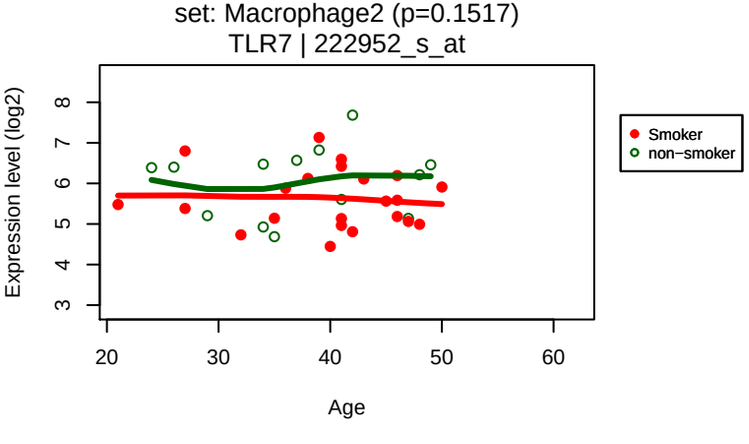

set: SmallAirway2 (p=0.5744)
TLR7 | 222952_s_at
Expression level (log2)
Age
Smoker
non−smoker

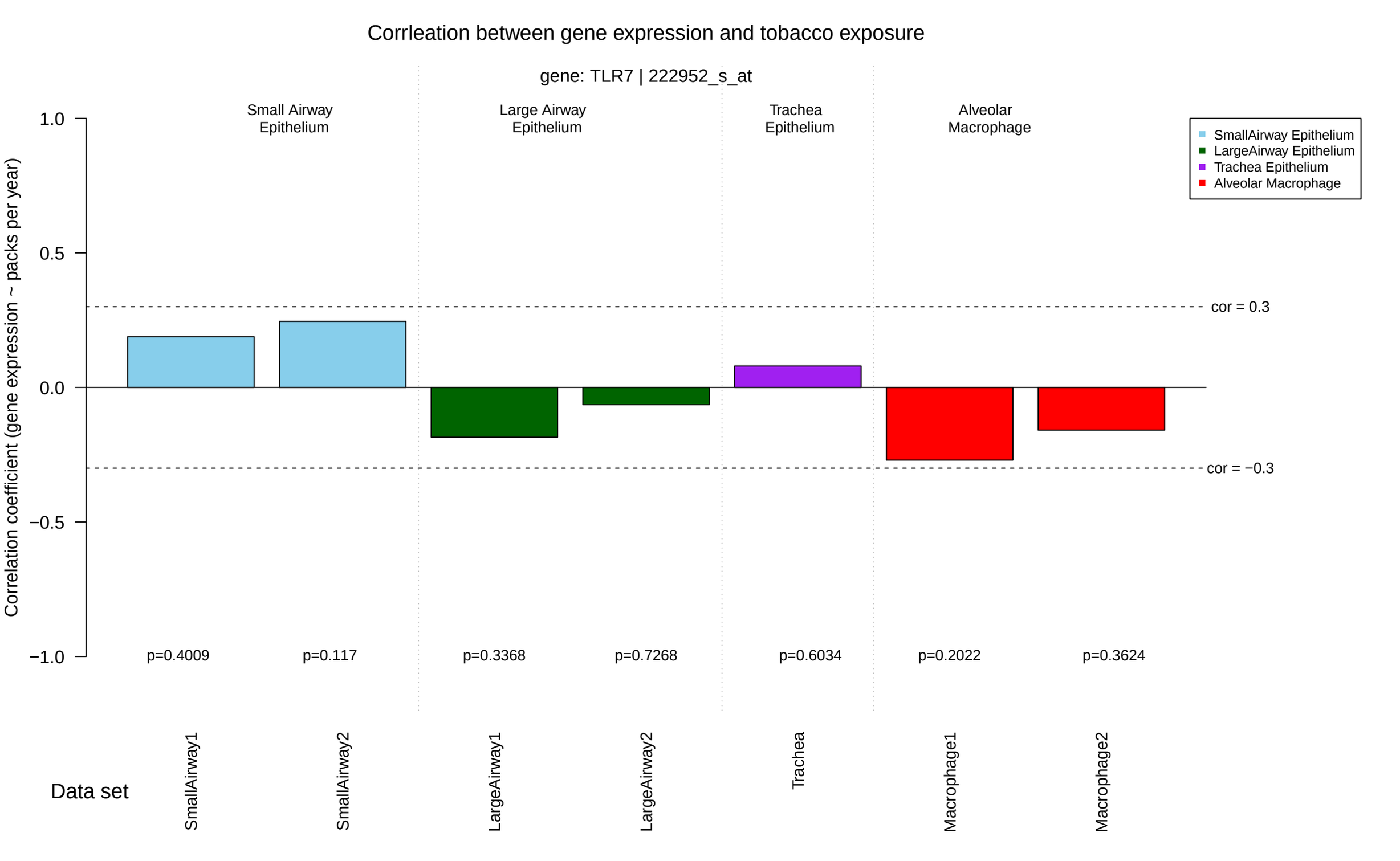

set: LargeAirway1 (p=0.5521)
TLR7 | 222952_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7155)
TLR7 | 222952_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2591)
TLR7 | 222952_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1647)
TLR7 | 222952_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1517)
TLR7 | 222952_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TLR7 | 222952_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4009
p=0.117
p=0.3368
p=0.7268
p=0.6034
p=0.2022
p=0.3624
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2